



THE AGE OF STREAM: AN ANALYSIS OF FILM TRENDS

Play Now



SOME BACKGROUND

Streaming platforms dominate the film industry.

This market is largely profitable, with more growth in sight.

How can production companies navigate this dynamic and saturated landscape?

\$84.7B → **\$112.7B**
IN 2024 BY 2029





OUR VISION

THE GOAL:

- The streaming market is saturated (Netflix, Prime Video, Disney+, etc.).
- We try to help production houses understand their content's position in this streaming market.

THE OBJECTIVE:

- To move beyond sheer quantity and provide an objective, data-driven analysis of content quality.
- Help producers and filmmakers understand where their audience is and the platform that will do justice to their content and get them a good prize for their movie.



OUR RESEARCH QUESTIONS

01 Across streaming services, how is content significantly affected by licensing, IMDb, TMDb, and catalog shape?

02 Which platform offers the optimal environment for a new to gain maximum visibility?

03 Where does the platform's content strategy suggest the highest demand for external acquisitions?





OUR DATA



1
NETFLIX
kaggle



2
DISNEY+
kaggle



3
AMAZON
PRIME VIDEO
kaggle



4
HBO
kaggle



5
HULU
kaggle



6
APPLE TV
kaggle



METHODOLOGY & DATA INTEGRATION

Data Cleaning

- Standardized titles, genres, and release years
- Converted IMDb scores & votes to numeric
- Removed duplicates using (title + year)

Core Metrics

- Platform catalog size & IMDb averages
- Release Year and popularity indicators
- Genre diversity & top genres per platform

Data Integration

- Combined 6 platform catalogs into one dataset
- Added platform flags and computed exclusive vs multi-platform
- Expanded multi-genre entries for genre-level analysis

Comparative Analysis

- Exclusive vs multi-platform rating differences
- Genre-based performance
- Correlation between IMDb rating and platform count

Trend & Forecasting

- Year-wise content trends
- Linear regression to estimate future rating trajectory



General Counts

```
df_test['platform exclusive'].value_counts()
```

```
platform exclusive
exclusive          17191
multi-platform      584
Name: count, dtype: int64
```

```
platforms = {
    "Hulu": hulu,
    "HBO": hbo,
    "Apple": apple,
    "Disney": disney,
    "Amazon": amazon,
    "Netflix": netflix
}
summary_df = basic_streaming_analysis(platforms)
```

===== BASIC PLATFORM COUNTS=====

Platform	Total Titles	NaN TMDb	NaN IMDb
Hulu	2398	160	166
HBO	3294	268	372
Apple	170	3	3
Disney	1535	109	427
Amazon	9871	2082	1021
Netflix	5850	311	482

```
# What is the highest rated movie on each platform?
```

```
df_exclusive = df_test[df_test['platform exclusive'] == 'exclusive']

highest Rated = df_exclusive.loc[df_exclusive.groupby('platform')['imdb_score'].idxmax()]

print(highest Rated[['platform', 'title', 'imdb_score']])
```

	platform	title	imdb_score
13279	Amazon Prime Video	pawankhind	9.9
17755	Apple	come from away	8.6
5021	Disney+	assembled: the making of hawkeye	8.8
14684	HBO	the shawshank redemption	9.3
17217	Hulu	parasite	8.5
833	Netflix	chhota bheem & krishna vs zimbara	9.1

```
# What movies on each platform were voted for the most on IMDb?
```

```
df_exclusive = df_test[df_test['platform exclusive'] == 'exclusive']

highest Rated = df_exclusive.loc[df_exclusive.groupby('platform')['imdb_votes'].idxmax()]

print(highest Rated[['platform', 'title', 'imdb_votes']])
```

	platform	title	imdb_votes
7322	Amazon Prime Video	the usual suspects	1059480.0
17743	Apple	coda	140450.0
4548	Disney+	the avengers	1353907.0
14960	HBO	the dark knight	2555504.0
16955	Hulu	deadpool	996056.0
89	Netflix	forrest gump	2021343.0



Terminology

TMDb Popularity:

favorites, votes, watchlist adds,
engagement

IMDb Rating:

average user voting behaviors on a
scale of 1-10

Production Country:

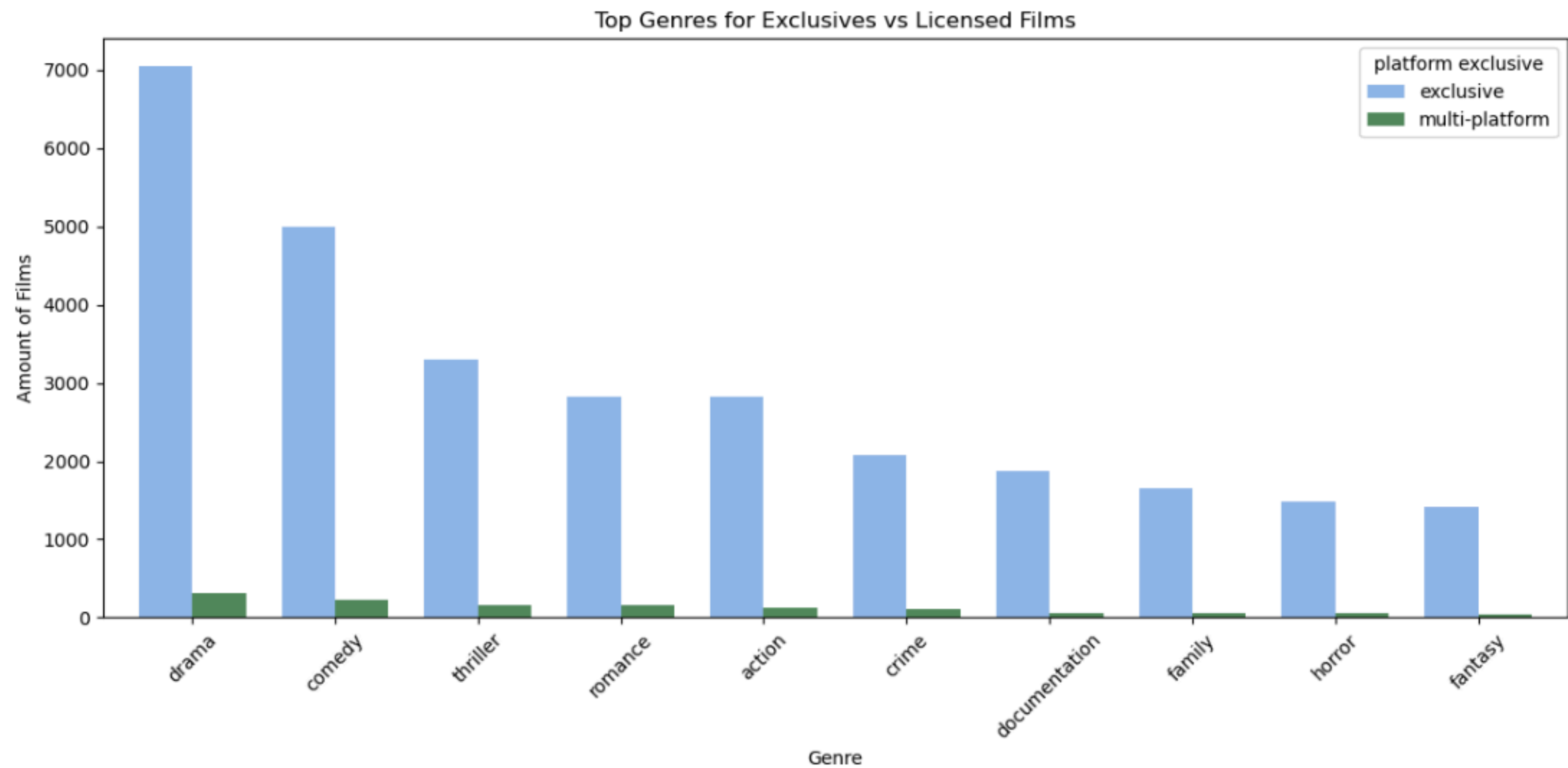
where the movie was funded and
manufactured

Exclusive/Licensed:

film is only on one platform versus
being on multiple

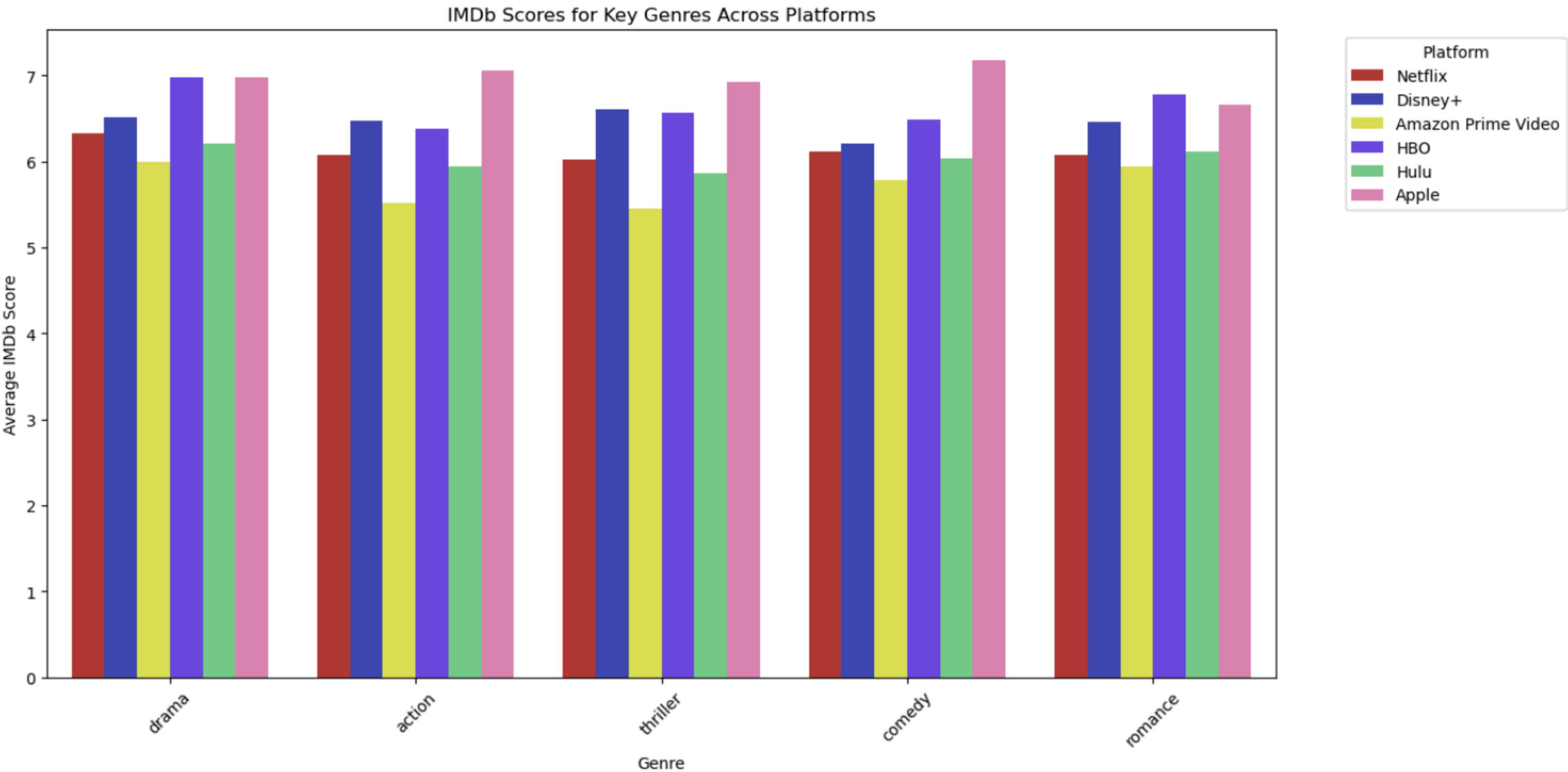


Genre at a Glance



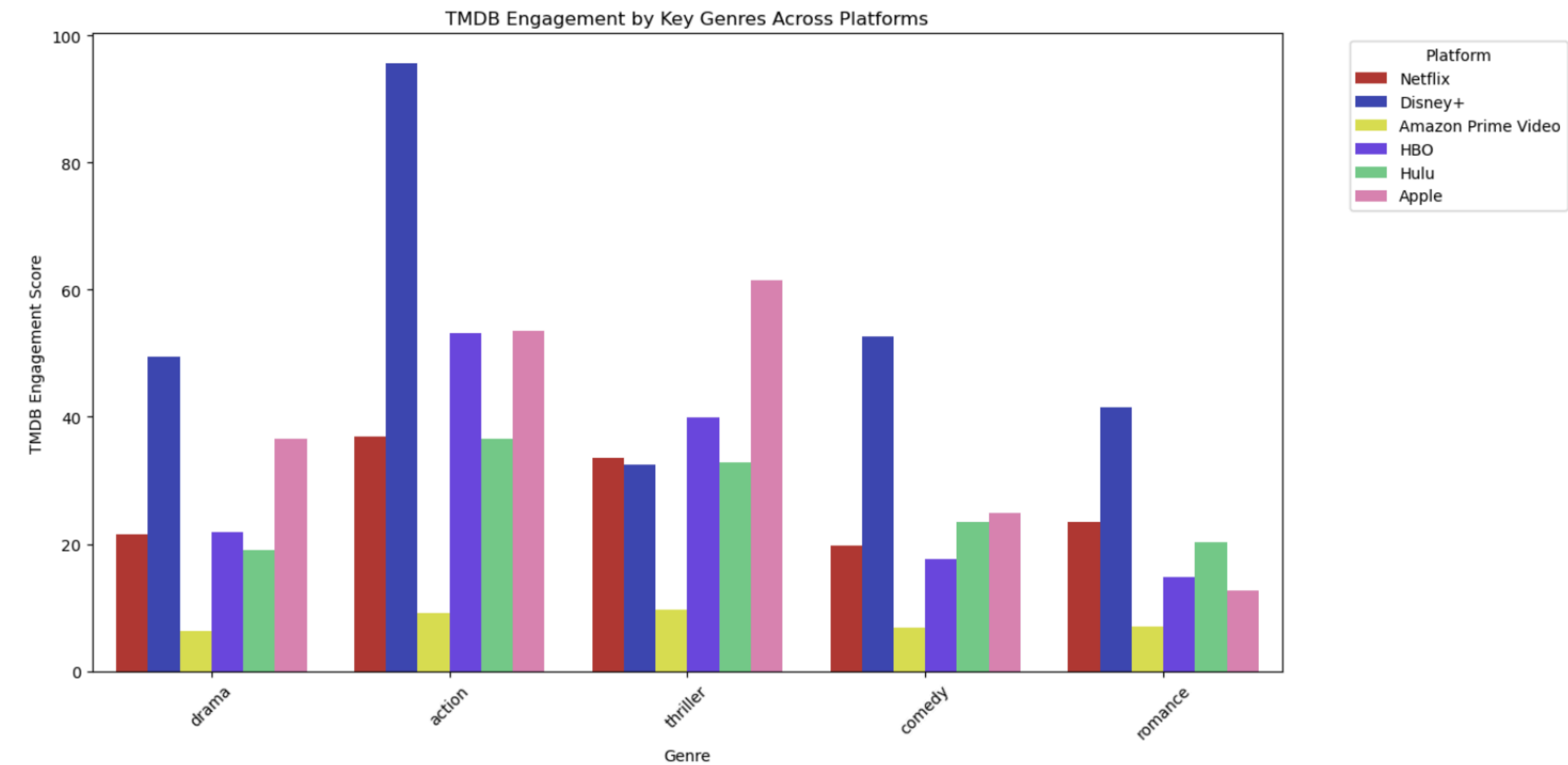


IMDb Scores by Genre and Platforms





Engagement by Genre Across Platforms





Production Countries at a Glance

- 1



UNITED STATES
- 2



INDIA
- 3



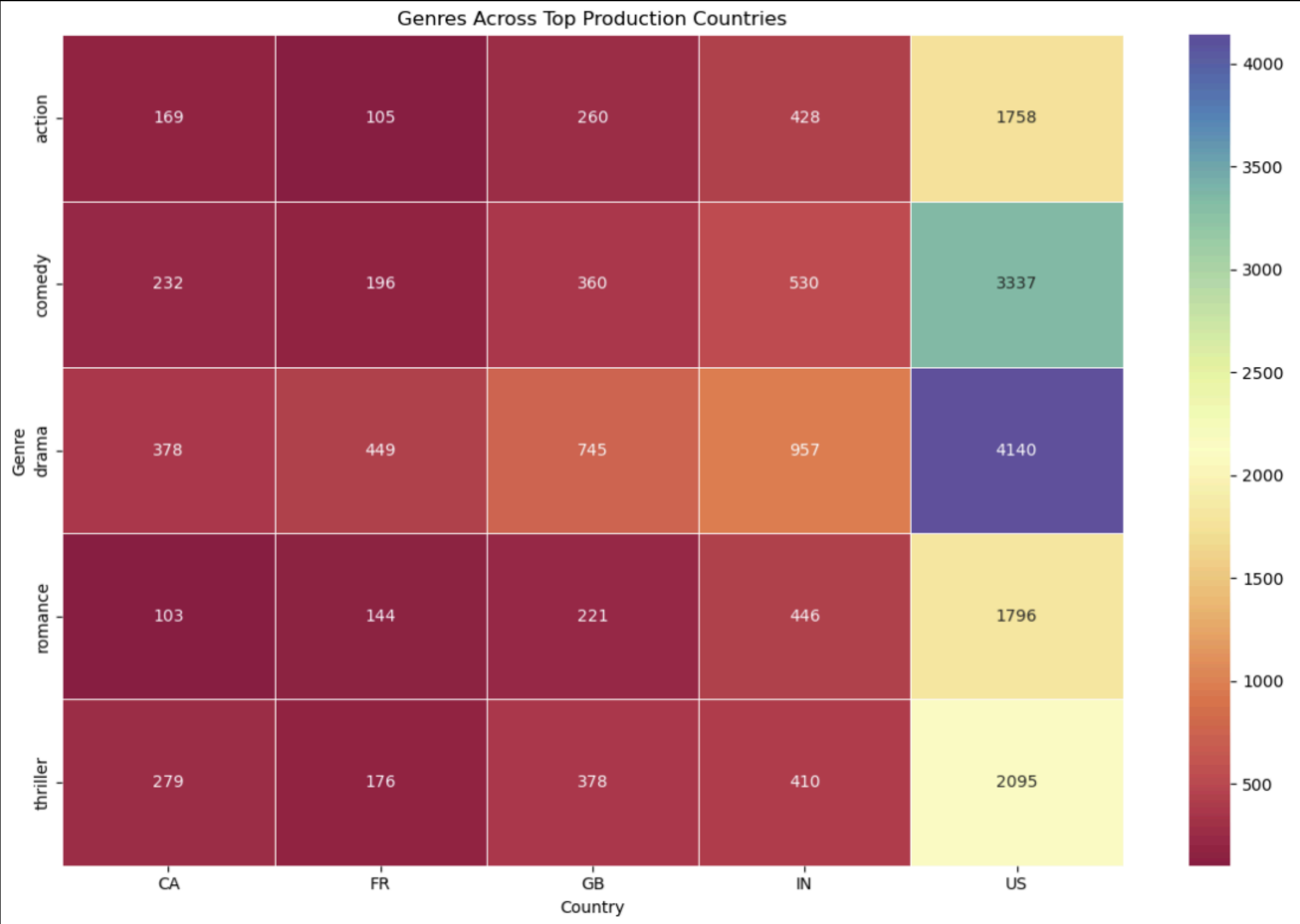
GREAT BRITAIN
- 4



CANADA
- 5

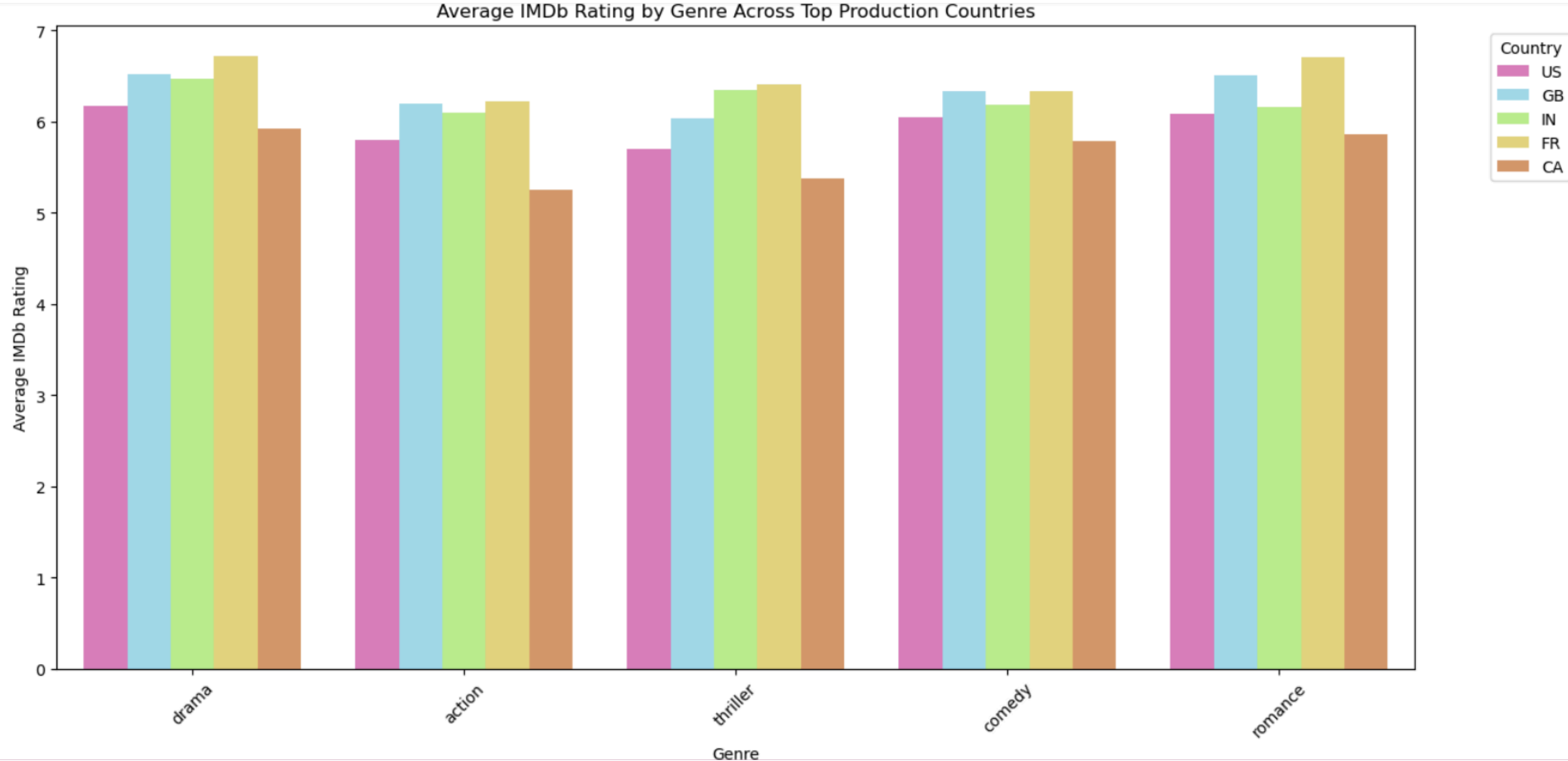


FRANCE

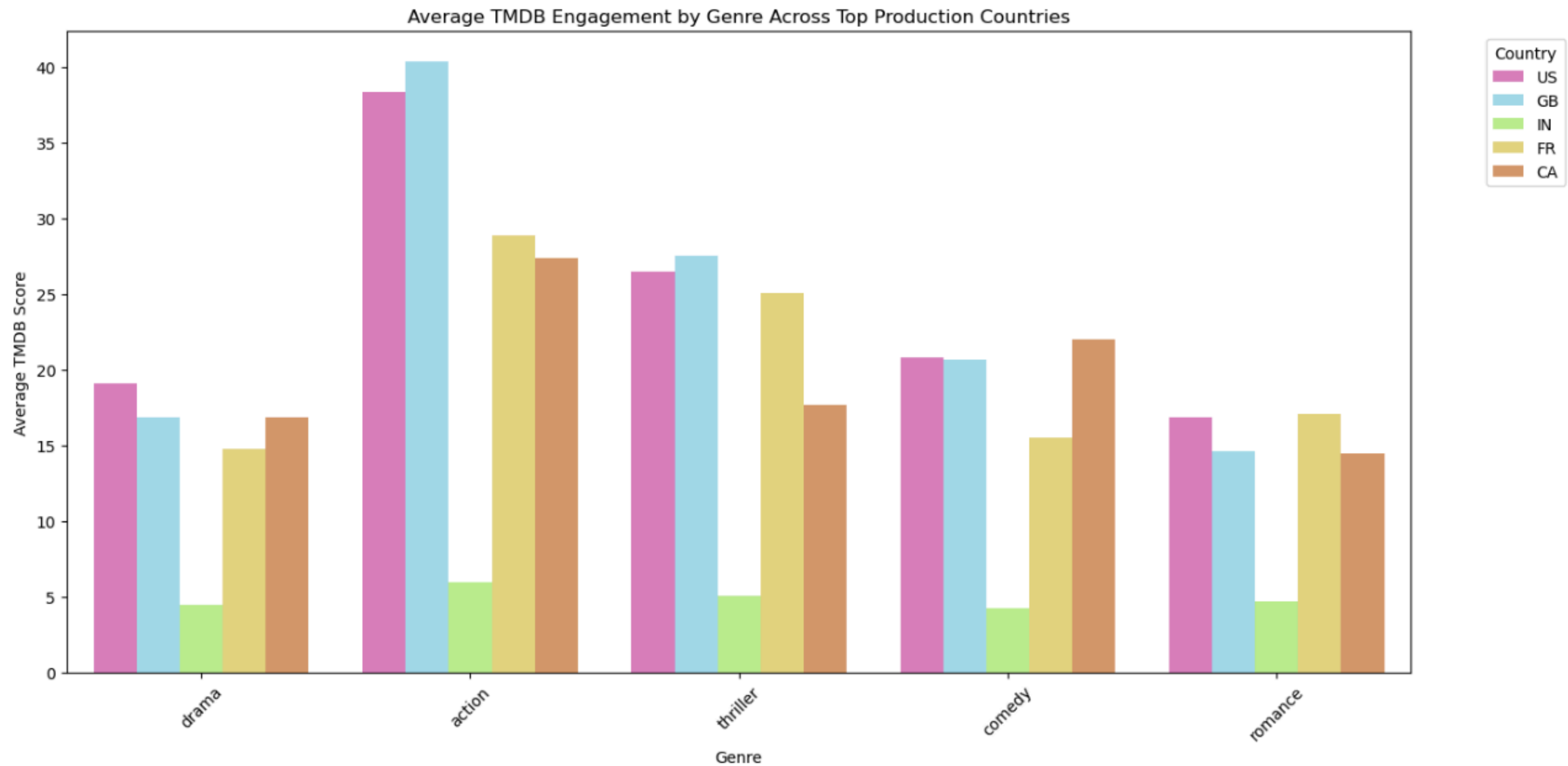




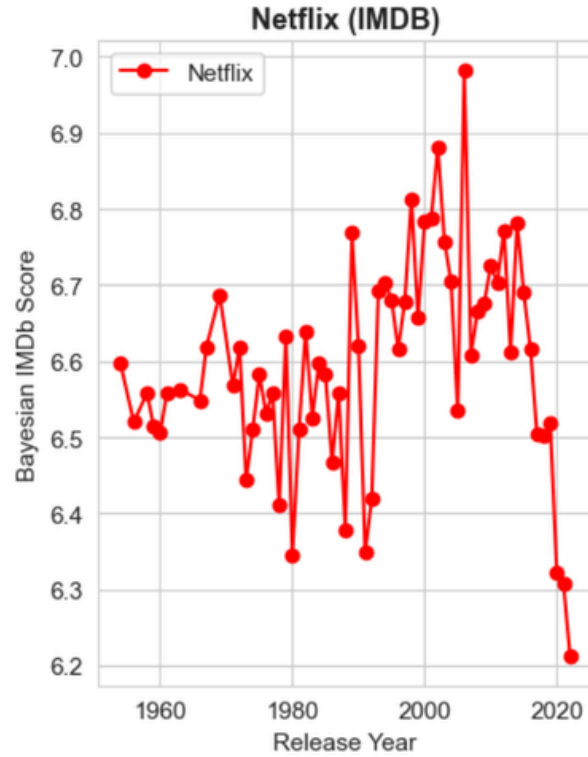
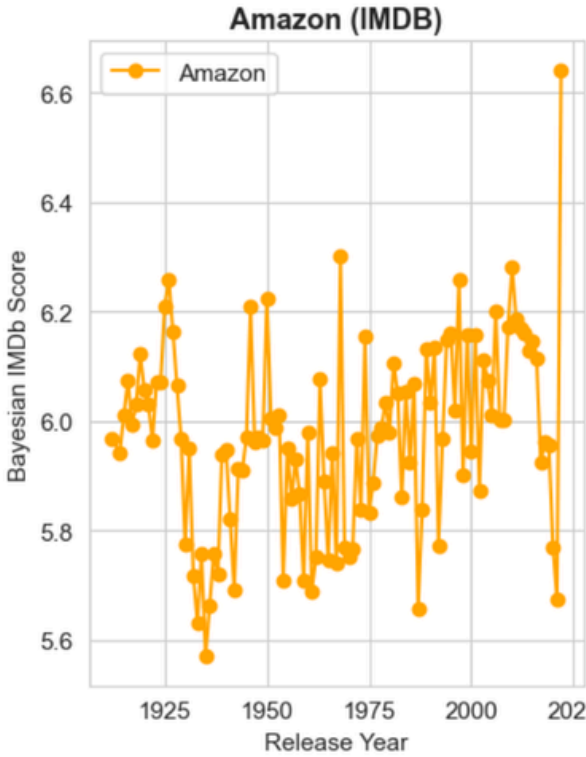
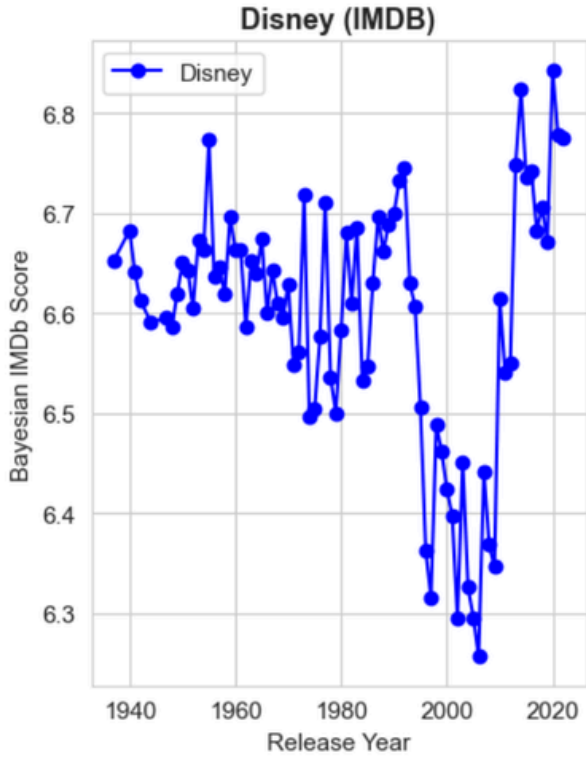
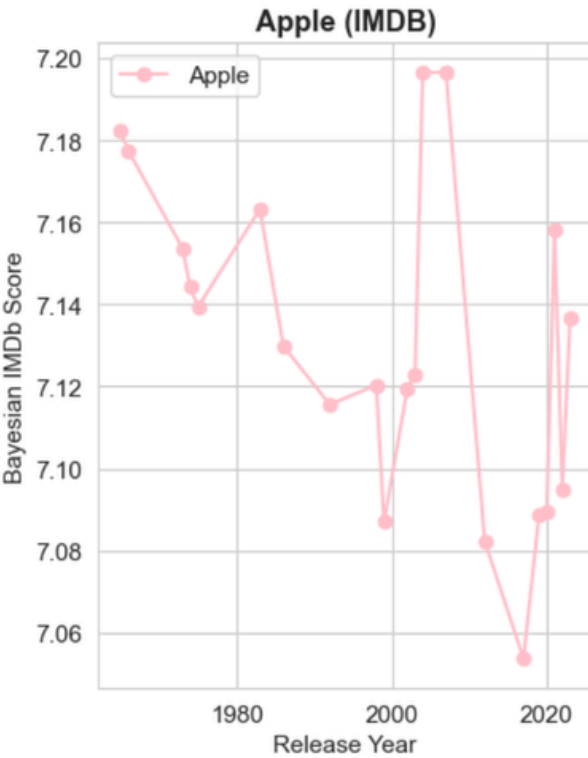
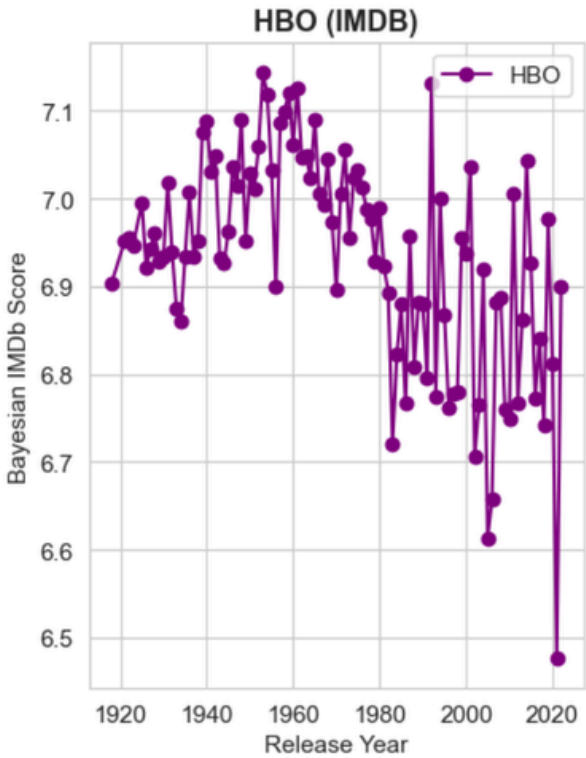
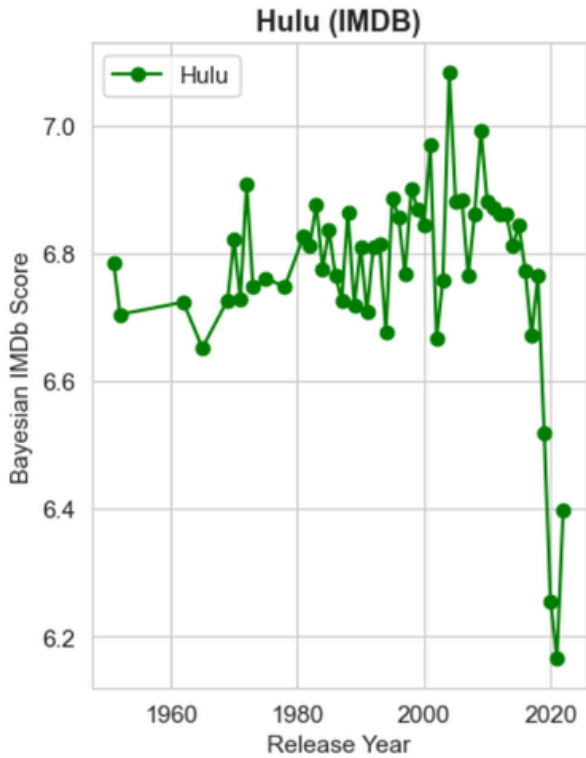
Origin of Production by User Ratings for Key Genres



Origin of Production by Engagement for key Genres

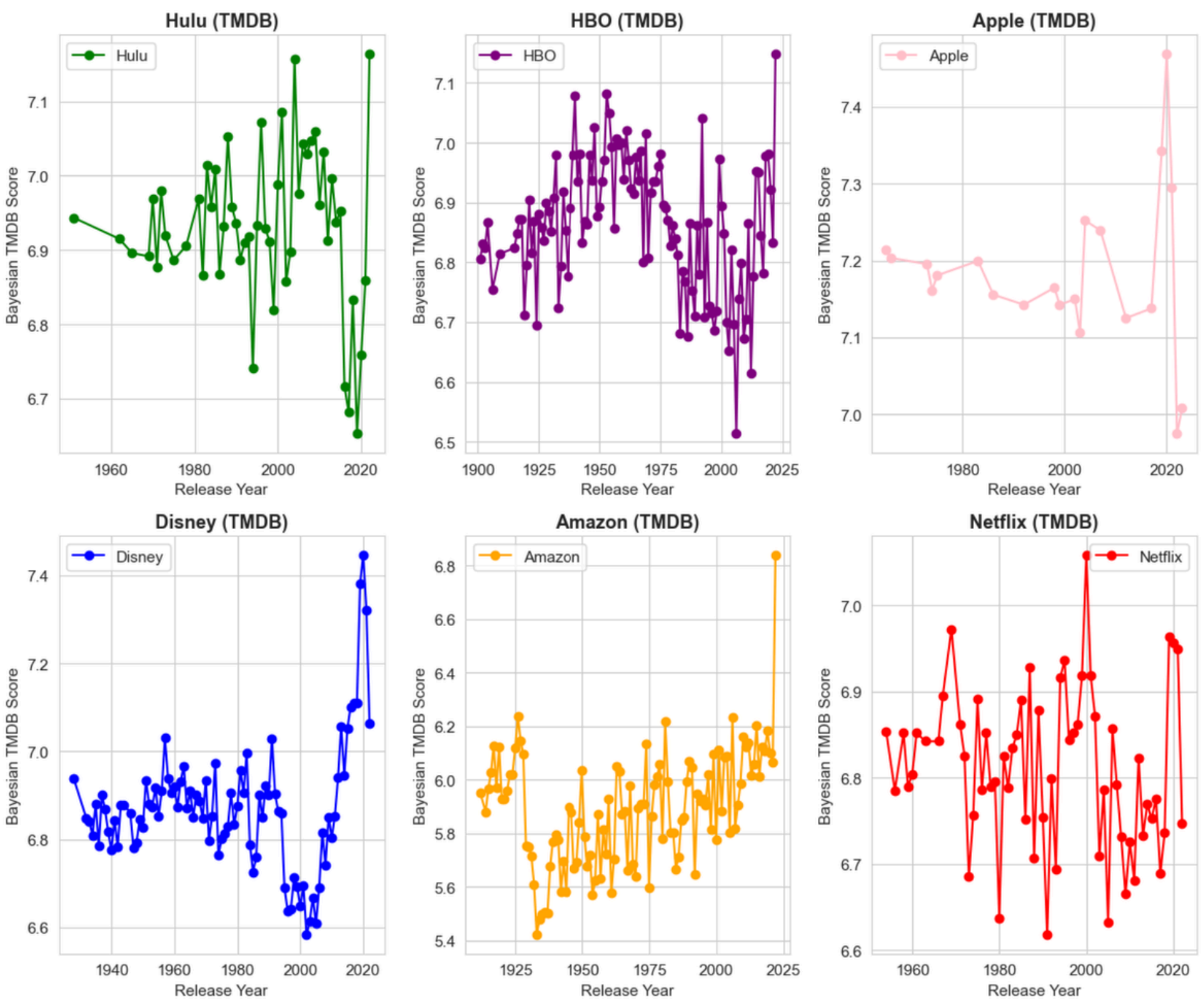


IMDB Score by Release Year





TMDB Score by Release Year



IMDB Ratings Based on Licensing Strategy

```
# Tests (research questions)

from scipy.stats import mannwhitneyu

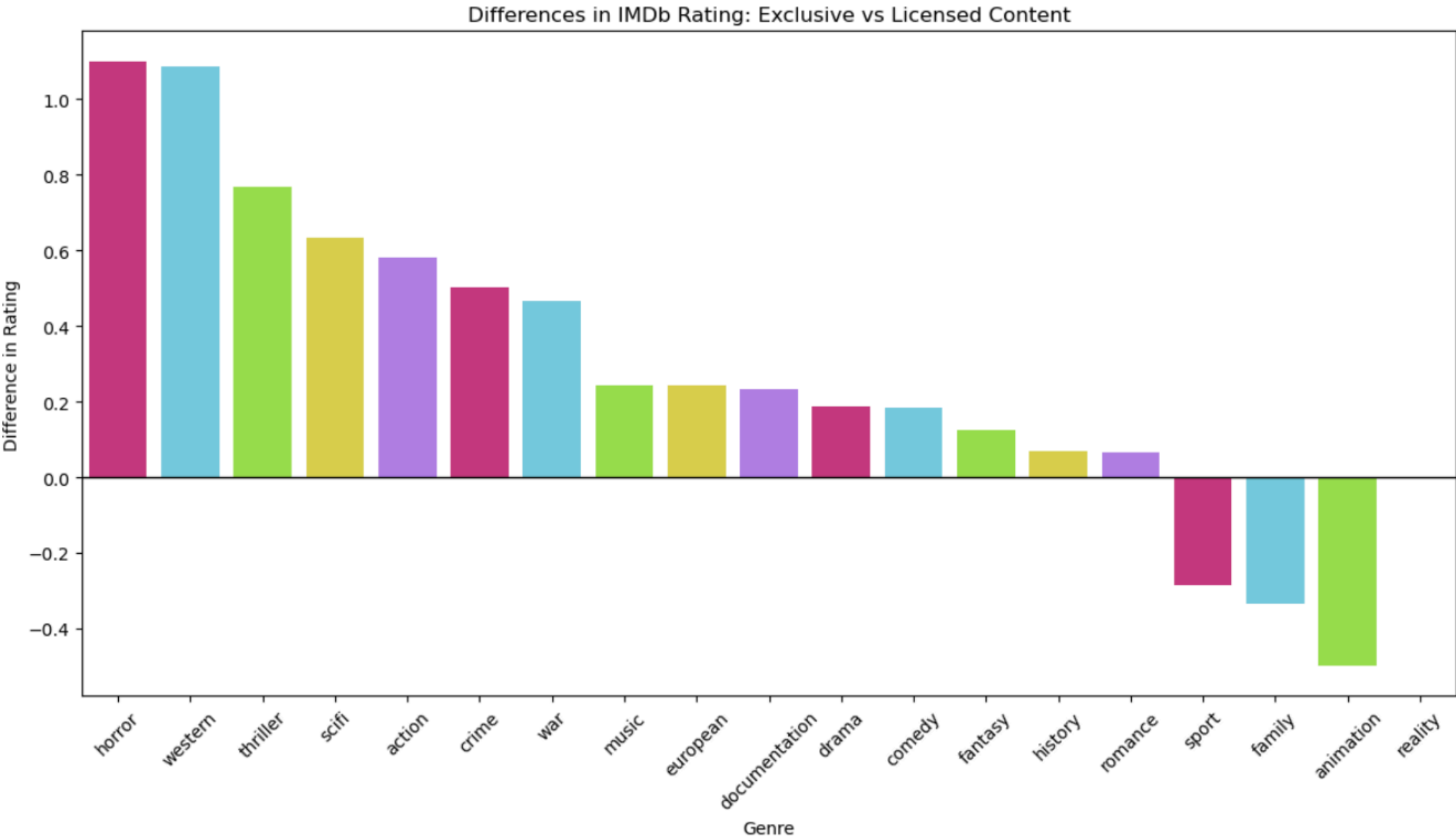
# Do exclusive movies have higher ratings than multi-paltform films?

df_test = df_test.dropna(subset=['imdb_score', 'platform exclusive'])

exclusive_rating = df_test[df_test['platform exclusive'] == 'exclusive']['imdb_score']
multi_rating = df_test[df_test['platform exclusive'] == 'multi-platform']['imdb_score']

stat, p = mannwhitneyu(exclusive_rating, multi_rating, alternative='two-sided')
print(f"U: {stat: .2f}, p-value: {p: .10f}")

U: 3109604.00, p-value: 0.0000000003
```



TMDB Ratings Based on Licensing Strategy

```
# Tests (research questions)

from scipy.stats import mannwhitneyu

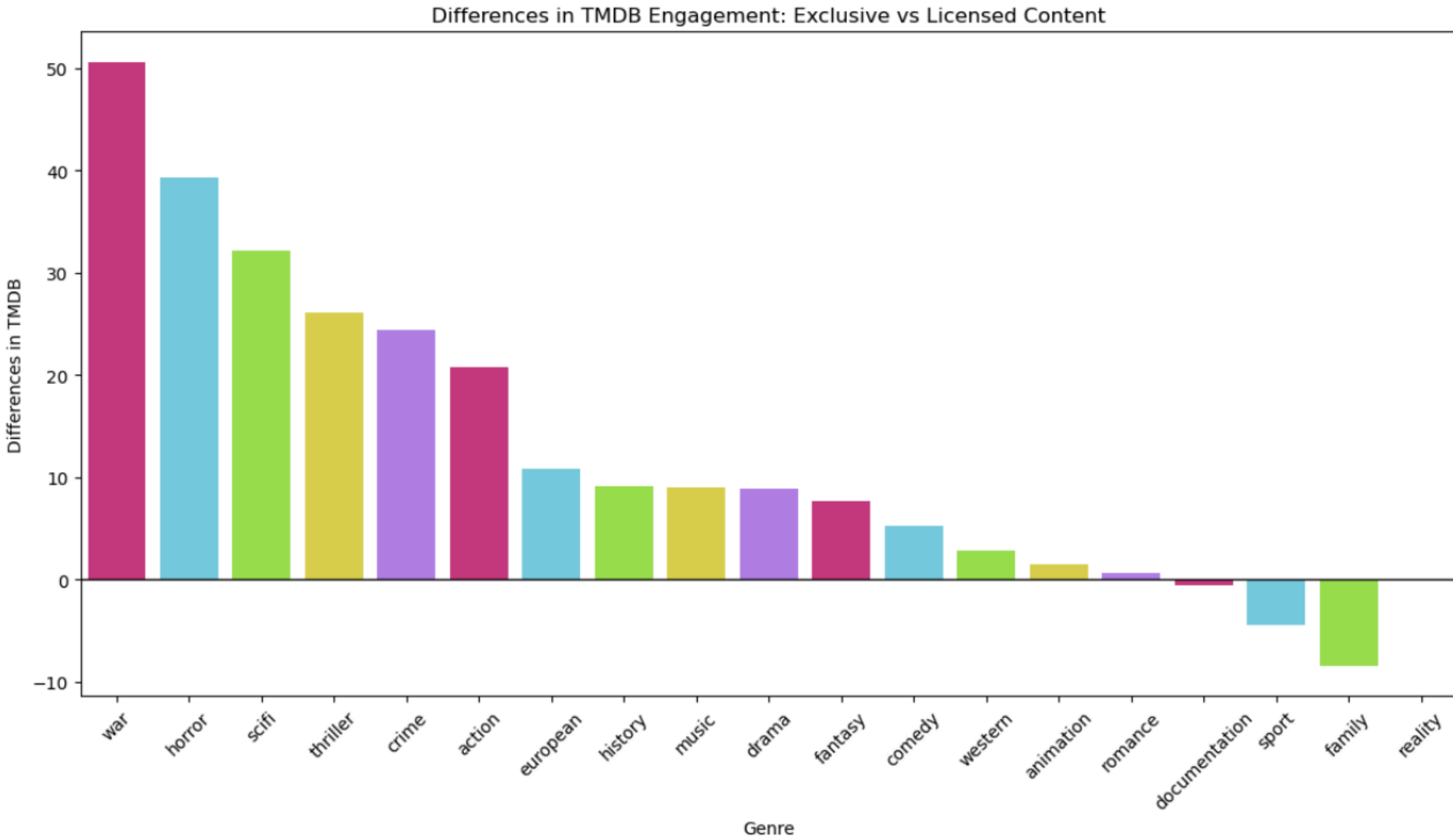
#Do exclusive movies have higher TMDB ratings than multi-paltform films?

df_test = df_test.dropna(subset=['tmdb_score', 'platform exclusive'])

exclusive_trating = df_test[df_test['platform exclusive'] == 'exclusive']['tmdb_score']
multi_trating = df_test[df_test['platform exclusive'] == 'multi-platform']['tmdb_score']

stat, p = mannwhitneyu(exclusive_trating, multi_trating, alternative='two-sided')
print(f"U: {stat: .2f}, p-value: {p: .10f}")

U: 3212657.00, p-value: 0.0000002217
```





CONCLUSION

Streaming Service Numbers: Quality vs. Quantity

Prime Video

Catalog Size: Largest overall (11,847 movies).

IMDb Rating: Lowest overall average (5.768).

Volume Insight: Hosts the highest absolute count of top-rated films (IMDb ≥ 8.0).

Hulu

IMDb Rating: Good-quality (Avg. IMDb 6.136).

Catalog Size: Mid-sized library (889 movies).

Genre Metric: Highest-rated genre is Documentary (Avg: 6.856).

Disney+

IMDb Rating: Highest overall average among the Top 5 set (6.441).

Catalog Size: Smallest catalog among the Top 5 set (563 movies).



CONCLUSION

Streaming Service Numbers: Quality vs. Quantity

HBO Max

IMDb Rating: Second highest percentage of content over 7.5 ($\approx 28.9\%$ total; 23.7% of its movies are over 7.5).

Volume Insight: Prioritizes quality density, offering a high probability of finding well-rated content.

Apple TV+

Catalog Size: Smallest library (hundreds of titles).

IMDb Rating: Highest average rating (reported ≈ 7.1 to 7.24) and the highest percentage of 7.5+ content ($\approx 38.3\%$).

Netflix

Catalog Size: Large library, estimated at around 3,800 movies in the US.

IMDb Rating: Second-highest quantity (Avg IMDb 6.259).



THANK YOU